CAPSTONE 1 — RETAIL SALES, MARGIN & INVENTORY ANALYTICS

Business Scenario: Retail Sales, Margin & Inventory Analytics Capstone

End-to-End BI Engineering Capstone (40 Hours)

Organization Background

A mid-sized multi-store retail chain operates across multiple regions, offering a diverse portfolio of consumer products. The organization has grown rapidly over the last few years, but its decision-making processes are still largely driven by static reports and manual spreadsheet analysis.

Business Problem

Senior leadership has identified declining profit margins despite steady sales growth.

There is limited visibility into:

- Which products truly drive profitability

- Which stores consistently underperform on margin

- How data quality issues impact reported KPIs

- How inventory and pricing decisions affect overall margin

Different departments (Sales, Finance, Operations) are reporting conflicting numbers due to inconsistent data definitions and lack of centralized analytics.

Objective of the Capstone

The objective of this capstone is to build a production-ready Retail Analytics BI solution that delivers:

- Accurate, reconciled Sales, Cost, and Margin KPIs

- Transparent handling of data quality issues

- A clean, auditable data model usable across BI tools

- Interactive dashboards that support executive and operational decision-making

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Scope of Analysis

The analytics solution will cover:

- Transaction-level retail sales analysis
- Product-level margin contribution
- Store and regional performance comparison
- Customer purchasing behavior
- Time-based trend analysis
- Scenario simulations for discounting and margin thresholds

Data Challenges (Realistic Business Constraints)

The raw retail data reflects real-world operational issues:

- Missing product, store, and customer references
- Duplicate transactions due to system sync failures
- Incorrect pricing and quantity anomalies
- Inconsistent date and numeric formats
- Intentional data quality injections to simulate production risks

All data cleansing and modeling must be performed strictly within BI tools to mirror enterprise BI constraints.

Expected Business Outcomes:

At the end of the engagement, stakeholders expect:

- A single version of truth for Sales and Margin KPIs
- Confidence in reported numbers through audit flags and reconciliation
- Identification of low-margin products and stores
- Actionable insights to optimize pricing, promotions, and inventory

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- A scalable BI framework reusable for future retail datasets

Success Criteria:

The project is considered successful when:

- KPIs are defensible across Power BI, Tableau, and Qlik Sense

- Dashboards support drill-down and root-cause analysis

- Data quality issues are visible, not hidden

- Business users can simulate scenarios and understand margin impact

- The solution can withstand executive and interview-level scrutiny

CAPSTONE CONSTRAINTS (MANDATORY TO BE FOLLOWED)

- **Tools only**
    - Power BI
    - Tableau
    - Qlik Sense
- ❌ No SQL engines
- ❌ No Python
- ❌ No external ETL / cloud tools
- All transformations must be done using:
    - Power Query
    - Tableau Prep
    - Qlik Load Script
- Raw datasets must remain unchanged

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TEAM & DURATION

- **Duration:** 5 Days × 8 Hours = **40 Hours**
- **Team: 6-7 Members – Below are the Tools to be used:**
    - 2 Power BI
    - 2 Tableau
    - 2 Qlik Sense

DATASETS Overview: (RETAIL)

Retail_Capstone1/

├── sales_transactions.csv (~6,000 rows)

├── products.csv (~200 rows)

├── stores.csv (~50 rows)

├── customers.csv (~1,000 rows)

├── calendar.csv (~60 days)

└── data_quality_injections.csv

All datasets are **intentionally imperfect** and must be fixed **inside BI tools only**.

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🟦 DAY 1 — DATA UNDERSTANDING & PROFILING (8 HOURS)

BI DATA ENGINEERING

Objective

Understand **raw retail data behavior**, risks, and KPI feasibility before any transformation.

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1. Business & KPI Alignment (1.5 hrs)

Generic Tasks

- Confirm KPIs:
  - Total Sales
  - Total Cost
  - Margin
  - Margin %
- Map KPIs → required columns
- Identify KPIs sensitive to bad data (null price, negative qty)

Tool Expectations

- Power BI: KPI logic drafted before DAX
- Tableau: KPI definitions before calculated fields
- Qlik: KPI logic before Set Analysis

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2. Raw Data Ingestion (2 hrs)

Generic Tasks

- Load all datasets as-is
- Preserve raw layer
- No cleansing or filtering

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Tool Expectations

- Power BI: RAW queries, disable load
- Tableau: Prep input steps only
- Qlik: RAW_* script sections

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3. Data Profiling & Diagnostics (3 hrs)

Generic Tasks

- Identify:
    - Null % per column
    - Mixed numeric/text fields
    - Invalid date formats
    - Outliers
    - Duplicate transaction IDs
- Document data risks

Tool Expectations

- Power BI: Column quality & distribution
- Tableau: Profiling pane
- Qlik: Script inspection + Model Viewer

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4. Grain & Join Validation (1.5 hrs)

Generic Tasks

- Validate fact grain:
    - 1 row = 1 product per transaction
- Validate joins:
    - Sales → Product
    - Sales → Store
    - Sales → Customer
- Identify need for UNKNOWN dimension rows

✅ **DAY 1 DONE IF**

✓ Raw data preserved

✓ Profiling documented

✓ Join & grain risks identified

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🟦 DAY 2 — DATA CLEANSING & STANDARDIZATION (8 HOURS)

BI DATA ENGINEERING

Objective - Create **curated retail datasets** that are clean, auditable, and reporting-safe.

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5. Data Type Standardization (2 hrs)

Generic Tasks

- Remove currency symbols & commas
- Convert numeric text → numbers
- Normalize date formats
- Flag parse failures

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6. Deduplication Logic (2 hrs)

Generic Tasks

- Identify duplicate transaction_id
- Keep latest row using created_ts
- Retain duplicate flag for audit

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7. Missing Value Handling (1.5 hrs)

Generic Tasks

- Define:
    - Drop vs flag vs impute
- Implement UNKNOWN strategy for:
    - product_id
    - store_id

- customer_id

8. Anomaly & Outlier Engineering (1.5 hrs)

Generic Tasks

- Flag:
  - Quantity ≤ 0
  - Negative sales
  - Extreme values (> P99 × 3)
- Parameterize thresholds

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9. Audit & Reusability (1 hr)

Generic Tasks

- Add DQ flags
- Add lineage fields
- Modularize logic
- Parameterize thresholds

✅ **DAY 2 DONE IF**

✓ Curated fact & dimensions created

✓ Deduplication applied

✓ Audit flags present

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🟦 DAY 2.5 — MODELING & SEMANTIC READINESS (4 HOURS)

BI DATA ENGINEERING (TRANSITION)

Objective

Prepare **BI-optimized star schema** for reporting.

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10. Dimensional Modeling (2 hrs)

Generic Tasks

- Build star schema:
    - FactSales
    - DimDate
    - DimProduct
    - DimStore
    - DimCustomer
- Validate cardinality
- Add UNKNOWN rows

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11. Semantic Readiness Checks (2 hrs)

Generic Tasks

- Hide technical fields
- Rename columns for business clarity
- Validate aggregation safety
- Test slicing behavior

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✅ **DATA ENGINEERING PHASE COMPLETE**

✓ Clean data

✓ Stable model

✓ Reporting-ready semantics

DAY 3 — KPI SEMANTICS & ANALYTICAL LOGIC (8 HOURS)

BI REPORTING / ANALYTICAL ENGINEERING

Objective

Implement **correct, defensible retail KPIs**.

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12. KPI Measure Engineering (3 hrs)

Generic Tasks

- Build:
    - Total Sales
    - Total Cost
    - Margin
    - Margin %
- Apply inclusion/exclusion rules
- Validate vs raw totals

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13. Context & Time Logic (2 hrs)

Generic Tasks

- Implement:
    - Daily / Monthly trends
    - Period comparisons
- Validate context behavior

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14. Parameter & Scenario Analysis (1.5 hrs)

Generic Tasks

- Create:
    - What-if discount scenarios
    - Margin threshold simulations
- Validate KPI impact

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15. KPI Reconciliation Engineering (1.5 hrs)

Generic Tasks

- Build visuals showing:
    - Included vs excluded rows
- Cross-tool KPI validation

✅ **DAY 3 DONE IF**

✓ KPIs correct

✓ Context controlled

✓ Reconciliation proven

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DAY 4 — ANALYTICAL DASHBOARDS & INTERACTIVITY (8 HOURS)

BI REPORTING ENGINEERING

Objective - Enable **retail analysis**, not static viewing.

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16. Analytical Dashboards (3 hrs)

Generic Tasks

- Build dashboards for:
    - Sales trends
    - Product contribution
    - Store performance
    - Regional comparison

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17. Interactivity Engineering (3 hrs)

Generic Tasks

- Implement:
    - Drill-down
    - Drillthrough
    - Tooltips
- Validate navigation

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Interactivity Engineering Continued:

1. **Hierarchical Drill-down**
    - Region → State → Store → Product → Transaction
2. **Product Performance Drillthrough**
    - Drillthrough from category KPI → product detail → transaction list
3. **Dynamic Top-N Selector**
    - Parameter to switch Top 5 / 10 / 20 products or stores
4. **Margin Sensitivity What-If**
    - Interactive margin % threshold slider impacting visuals
5. **Cross-Highlighting Between KPIs**
    - Selecting a store highlights margin, discount, and returns simultaneously
6. **Promo Impact Toggle**
    - Toggle to include/exclude promo transactions in KPIs
7. **Custom Tooltip Pages**
    - Hover on product shows sales, margin trend, discount behavior
8. **Time-Window Comparison Selector**
    - MTD / QTD / YTD switch without duplicating visuals
9. **Outlier Visibility Toggle**
    - Show/hide extreme sales or margin outliers interactively
10. **Data Quality Overlay**
    - Toggle to visually flag transactions excluded from KPIs

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18. Performance-Driven Reporting (2 hrs)

Generic Tasks

- Measure load & interaction time
- Optimize visuals
- Reduce query load

✅ **DAY 4 DONE IF**

✓ Dashboards analytical

✓ Interactivity correct

✓ Performance optimized

🟩 DAY 5 — BI PRODUCTIZATION & STORYTELLING (4 HOURS)

BI REPORTING ENGINEERING

Objective

Deliver a **production-grade retail BI product**.

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19. BI Product Hardening (1.5 hrs)

Generic Tasks

- Remove unused visuals
- Lock KPIs
- Final validation

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20. Storytelling & Demo (2.5 hrs)

Generic Tasks

- Create narrative:

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  - Business problem
  - Insight
  - Action
- Prepare executive walkthrough
- Prepare defense Q&A

✅ **FINAL DONE IF**

✓ Retail BI solution production-ready

✓ Balanced data & reporting engineering

✓ Interview / client defensible

Conclusion

The Retail capstone successfully demonstrates how **transaction-level sales data**, often plagued with duplicates, pricing anomalies, and missing references, can be engineered into a **reliable analytical foundation** for business decision-making.

Through rigorous data profiling, cleansing, and star-schema modeling, learners ensured that **revenue, cost, and margin KPIs** were both **accurate and auditable**. Advanced semantic logic and parameterized analysis enabled meaningful insights into **product performance, store efficiency, and regional trends**.

This capstone highlights how BI Engineers play a critical role in **protecting financial KPIs from data quality risks**, ensuring that executive decisions are based on **correct margins, not misleading totals**.